Electrostatics

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Coulomb's Law & Electric Field

Source charges — responsible for creating electric field/force

Test charge — charge used to measure the exisiting electric field/force

Principal of Superposition — interaction between two charges is independant of other charges

Coulomb's Law

$$\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} \tag{1}$$